IT4010 Research Project — 2026 Jan

Bi-weekly Progress Log Entry

Section A — Student Information

Student Name	Rasara S.H.K
Registration No.	IT21313998
Project Title	An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka
Supervisor	Ms. Tharushi Rubasinghe
Co-Supervisor	Ms. Sameeri Subasinghe
Research Cluster	TIM (Technology Integration and Management) / CoEAI
Specialization	Information Technology (IT)
Submission Period	22nd March - 28th March 2026
Date Submitted	28th March 2026

Section B — Progress Summary

Current Project Stage	Post-Proposal / Planning
Overall Status	On Track
% Completion (Est.)	15%

Section C — Tasks This Period

C1. Tasks Completed

- Documented panel feedback and suggestions
- Incorporated minor feedback into proposal document
- Began detailed literature review for implementation

C2. Tasks Planned for Next Period

- Update proposal document with panel feedback
- Begin detailed literature review on IoT sensors
- Research component specifications for rain gauge nodes
- Start system architecture design refinement

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

Challenge	Mitigation
• Panel question about API data availability "Ditwah" R package	Explained backup plan using CRAN
• Concern about field deployment permissions Week 10	Will obtain landowner permissions before

Section E — Supervisor Meeting Notes

Date of Meeting	28th March 2026
Mode	Online

Key Points Discussed

- Congratulations on successful presentation
- Reviewed panel feedback
- Discussed next steps for implementation phase

Action Items Assigned

- Update proposal document by 25th March
- Begin component procurement research
- Set up development environment

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: 

Name: Rasara S H K

Date: 28th March 2026

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____

Name: Ms. Tharushi Rubasinghe

Date: 28th March 2026

IT4010 Research Project — 2026 Jan

Bi-weekly Progress Log Entry

Section A — Student Information

Student Name	Rasara S.H.K
Registration No.	IT21313998
Project Title	An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka
Supervisor	Ms. Tharushi Rubasinghe
Co-Supervisor	Ms. Sameeri Subasinghe
Research Cluster	TIM (Technology Integration and Management) / CoEAI
Specialization	Information Technology (IT)
Submission Period	29th March – 4th April 2026
Date Submitted	4th April 2026

Section B — Progress Summary

Current Project Stage	Planning / Design
Overall Status	On Track
% Completion (Est.)	18%

Section C — Tasks This Period

C1. Tasks Completed

- Updated proposal document with panel feedback
- Added detailed data collection methodology
- Included API contingency plan
- Clarified integration points with Student B
- Added risk assessment for field deployment
- Began literature review on IoT sensor hardware

C2. Tasks Planned for Next Period

- Complete literature review on IoT sensors
- Research component suppliers and pricing
- Begin requirements gathering for sensor nodes
- Start system architecture design

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

Challenge	Mitigation
• Finding detailed specifications for tipping bucket rain gauges	Contacted 3 suppliers for technical datasheets

Section E — Supervisor Meeting Notes

Date of Meeting	No formal meeting
Mode	-

Key Points Discussed
Action Items Assigned

Student Declaration	**Supervisor Acknowledgement**
I confirm that this log entry is a true record of my research progress.	*I confirm I have reviewed this bi-weekly progress submission.*
Signature: 	Signature: _____
Name: Rasara S H K	Name: Ms. Tharushi Rubasinghe
Date: 4th April 2026	Date: 4th April 2026

IT4010 Research Project — 2026 Jan

Bi-weekly Progress Log Entry

Section A — Student Information

Student Name	Rasara S.H.K
Registration No.	IT21313998
Project Title	An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka
Supervisor	Ms. Tharushi Rubasinghe
Co-Supervisor	Ms. Sameeri Subasinghe
Research Cluster	TIM (Technology Integration and Management) / CoEAI
Specialization	Information Technology (IT)
Submission Period	5th -11th April 2026
Date Submitted	11th April 2026

Section B — Progress Summary

Current Project Stage	Planning / Design
Overall Status	On Track
% Completion (Est.)	20%

Section C — Tasks This Period

C1. Tasks Completed

- Completed literature review on IoT sensor hardware
- Researched ESP32 specifications and alternatives
- Researched LoRa module options (RFM95W selected)
- Researched tipping bucket rain gauge options
- Created supplier comparison sheet (5 suppliers)
- Began requirements gathering document

C2. Tasks Planned for Next Period

- Finalize system architecture design
- Begin component procurement
- Set up development environment

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

Section E — Supervisor Meeting Notes

Date of Meeting	No formal meeting
Mode	-

Key Points Discussed
Action Items Assigned

Student Declaration	**Supervisor Acknowledgement**
I confirm that this log entry is a true record of my research progress.	*I confirm I have reviewed this bi-weekly progress submission.*
Signature: 	Signature: _____
Name: Rasara S H K	Name: Ms. Tharushi Rubasinghe
Date: 11th April 2026	Date: 11th April 2026

IT4010 Research Project — 2026 Jan

Bi-weekly Progress Log Entry

Section A — Student Information

Student Name	Rasara S.H.K
Registration No.	IT21313998
Project Title	An Intelligent IoT-Based Flood Early Warning and Water Level Prediction System for Kelani Riverine Communities in Sri Lanka
Supervisor	Ms. Tharushi Rubasinghe
Co-Supervisor	Ms. Sameeri Subasinghe
Research Cluster	TIM (Technology Integration and Management) / CoEAI
Specialization	Information Technology (IT)
Submission Period	12th -18th April 2026
Date Submitted	18th April 2026

Section B — Progress Summary

Current Project Stage	Design / Requirements
Overall Status	On Track
% Completion (Est.)	22%

Section C — Tasks This Period

C1. Tasks Completed

- Completed functional requirements for IoT sensor nodes
- Completed non-functional requirements (performance, security)
- Began system architecture diagram
- Submitted data access request to Irrigation Department
- Prepared detailed budget estimation

C2. Tasks Planned for Next Period

- Finalize system architecture design
- Begin component procurement
- Set up development environment

Section D — Challenges & Mitigation

Describe any issues or blockers encountered and how you plan to address them.

Challenge	Mitigation
• Budget exceeded initial estimate	Identified cost-saving measures; revised budget

Section E — Supervisor Meeting Notes

Date of Meeting	No formal meeting
Mode	-

Key Points Discussed
Action Items Assigned

Student Declaration

I confirm that this log entry is a true record of my research progress.

Signature: 

Name: Rasara S H K

Date: 18th April 2026

Supervisor Acknowledgement

I confirm I have reviewed this bi-weekly progress submission.

Signature: _____

Name: Ms. Tharushi Rubasinghe

Date: 18th April 2026